EXAM

Inorganic Chemistry

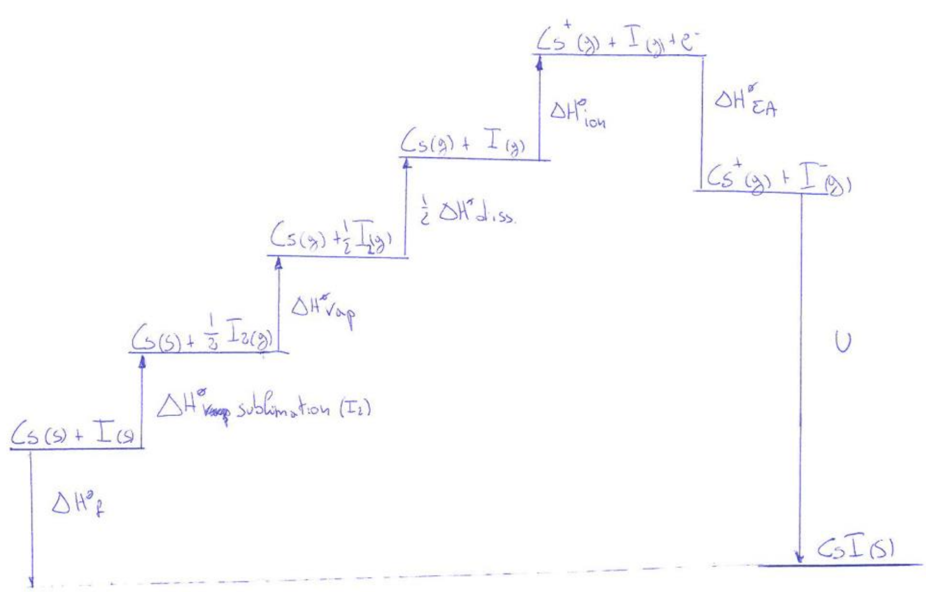
March 2012

1) By taking in consideration the position of the elements in the periodic table, Indicate with a cross the true sentences (6pt.):

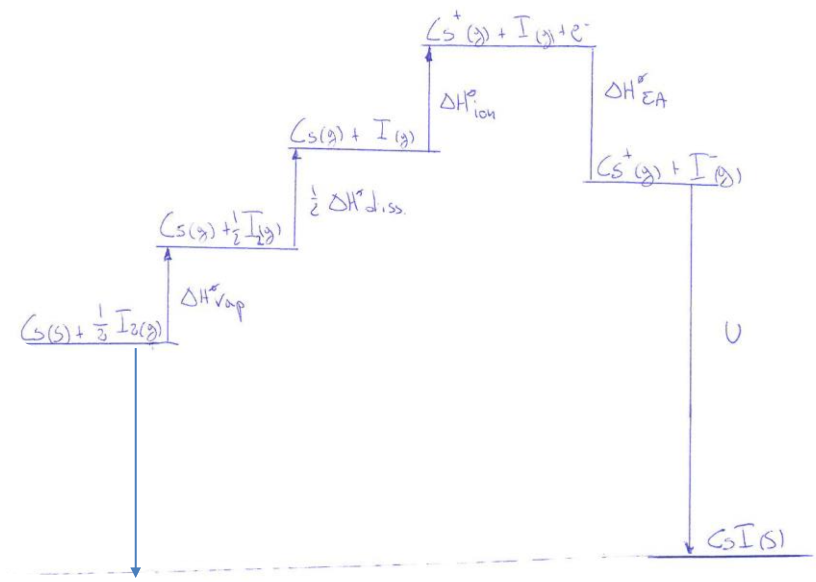
No In absolute values, iodine has a higher electron affinity than chlorine.
☑ Rubidium has a larger atomic radius than strontium
☑ Calcium has a higher first ionization energy than potassium
No Calcium has a higher second ionization energy than potassium
No The first ionization energy is higher for potassium than for lithium
No Sodium is more electronegative than fluorine.

2) Represent the Born-Haber cycle for the formation of cesium iodide (CsI), 7pt. Which forces have to be taken into account when we calculate the lattice energy of an ionic solid (3pt.)?

Two possible answers are accepted as correct:
a) At standard conditions iodine is a solid, and thus the sublimation temperature should be taken in consideration.



b) **Despite that, also the following answer will be considered correct in this exam, since it is the direct derivation of our rationale on the B-H cycle of the NaCl:**



Which forces have to be taken into account when we calculate the lattice energy of an ionic solid?
1) Coulombic attractive and repulsive energies, which account about the 90% of the total lattice energy;
2) Additional repulsive energy resulting from repulsion between the overlapping outer electron density of adjacent ions;
3) Other minor contributions as van der Waals energy

3) How can we calculate the effective nuclear charge according to Slater (2pt.)? Using Slater's rules, calculate a value for the effective nuclear charge felt by an electron in a *2p* orbital of a neutral Neon atom, 3pt. What are the limitations of the Slater's approach? (1pt.)

According to Slaters the effective nuclear charge (Z*) can be calculated with the following equation: $Z^* = \sigma Z$, where σ is a screening constant ad depend on the electronic configuration of the atom, and Z is the atomic number of the atom.

The electronic configuration of Ne is: $1s^2\ 2s^2\ 2p^6$

Thus $\sigma = 2 \times 0.85 + 7 \times 0.35 = 4.15$

Slater's rules are based on a very rough model, which doesn't take in consideration, for instance, the difference in charge densities between s and p orbitals, or between d and f orbitals.

4) Use the most appropriate definition to identify acids and bases in the following reactions (please, pay attention to the specific reaction environment, 8pt).

   a) $SiO_2 + CaO \rightarrow CaSiO_3$, is CaO a base or an acid? Why?

   CaO is a base, because it can donate an oxygen atom to silicon (Lux-Flood definition).

   b) $HNO_3 + H_2O \rightarrow NO_3^- + H_3O^+$ is H_2O a base or an acid? Why?

   H_2O is a proton acceptor, thus in this case is a base (Brønsted definition).

   c) $NH_3 + BF_3 \rightarrow H_3N–BF_3$ is NH_3 a base or an acid? Why?

   N uses a lone pare to form a N-B bond, thus NH_3 is a base (Lewis definition)

   d) $H_2N\underset{\overset{\|}{O}}{C}NH_2 + NH_3\ \text{(solvent)} \leftrightarrow H_2N\underset{\overset{\|}{O}}{C}NH^- + NH_4^+$ is $H_2N\underset{\overset{\|}{O}}{C}NH_2$ a base or an acid? Why?

   $H_2NC(O)NH_2$ increases the concentration of the solvent cationic species, thus it is an acid accordin to the solvent system definition.

5) Determine the oxidation state and the coordination number of the central atom for each of the following complexes, 4pt:

   e) $[Co(NH_3)_4(H_2O)Cl]Cl$,

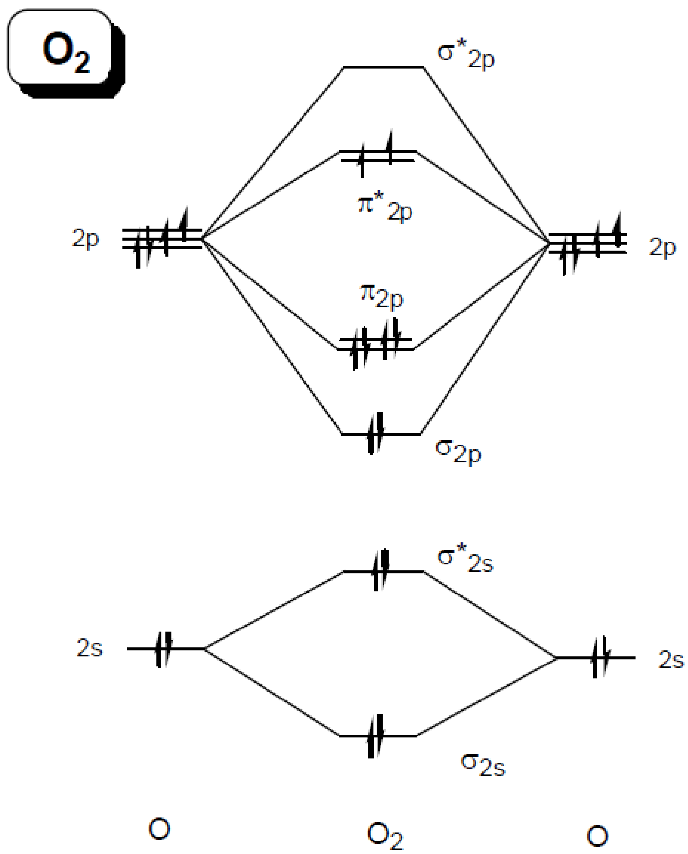
   f) $K_2[Zn(OH)_4]$

   Suggests possible geometries (linear, cubic, octahedral, …) for these two compounds, 2 pt.

compound	**Ox. Number**	**Coord. Number**	**Geometry**
$[Co(NH_3)_4(H_2O)Cl]Cl$	**2**	**6**	**The most probable geometry is the octahedron. (Distortions of the octahedron are also possible, and the trigonal prism has been also**

			observed for coordination number 6, but rarely
$K_2[Zn(OH)_4]$	**2**	**4**	**For coordination number 4 ,the most common geometry is the tetrahedron, but also square planar conformation is quite present.**

6) Use the MO theory to explain the bonding in O_2 (6pt.). Use your diagram to predict the bond order in O_2 molecules (2pt.). Do you think O_2 is a paramagnetic substance? Why (2pt.)?



There are 8 electrons in bonding orbitals and 4 electrons in anti-bonding orbitals, thus (8-4)/2=2 the bonding order is 2

Oxygen is expected to be paramagnetic since it has 2 unpaired electrons.

7) What are the most important properties for a solvent? (4 pt.)

a) **The temperature-pressure range over which it is a liquid.**
b) **Its dielectric constant.**
c) **Its donor and acceptor properties: solvent polarity**
d) **Its prontonic (Brønsted) acidity or basicity.**
e) **The nature and extent of autodissociation.**